

Module 04: Cognition — Decision-Making and Prefrontal Function

Course: Active Inference 401 | **Unit:** Neuroscientific Frontiers | **Audience:**
Advanced undergraduates / graduate students

Learning Objectives

1. Analyze how **prefrontal cortex** implements hierarchical planning and policy evaluation in Active Inference.
2. Evaluate the neuroscience of **expected free energy** — how the brain decomposes decisions into pragmatic and epistemic components.
3. Examine the role of **dopaminergic circuits** in policy precision and confidence.

Key Concepts

1. Prefrontal Cortex as Hierarchical Policy Evaluator

The prefrontal cortex (PFC) implements the highest levels of the policy hierarchy — evaluating multi-step plans and selecting among competing strategies:

Dorsolateral PFC (dlPFC): Maintains working memory representations of candidate policies — the "workspace" where alternative action sequences are held and compared. fMRI studies show dlPFC activation scales with the number of policies being evaluated and the depth of planning required.

Ventromedial PFC (vmPFC): Computes expected value — estimating how much reward (pragmatic value) each policy will yield. The vmPFC integrates information about past outcomes (from hippocampus) and current state estimates (from sensory cortex) to compute predictive value signals.

Anterior cingulate cortex (ACC): Monitors for policy conflicts and tracks prediction errors about expected difficulty. The ACC signal is strongest when multiple policies have similar

expected value (high uncertainty about which policy to select) — this is the neural signature of **decision difficulty**, formalized as low precision of the posterior over policies.

Frontopolar cortex (BA10): Implements **counterfactual reasoning** — evaluating policies that are *not* currently being executed. This region is critical for maintaining an alternative strategy "in the background" and switching when the current strategy begins to fail. This corresponds to the deep temporal models of Active Inference, where multiple policies are evaluated simultaneously over long time horizons.

2. Neural Decomposition of Expected Free Energy

Expected free energy (EFE) = pragmatic value + epistemic value. These components are neurally dissociable:

Pragmatic value (reward-seeking): Encoded by vmPFC and ventral striatum. Dopaminergic signals from VTA to ventral striatum encode the expected reward associated with each policy. The orbitofrontal cortex tracks the subjective value of expected outcomes.

Epistemic value (information-seeking): Encoded by dorsal ACC, anterior insula, and frontopolar cortex. These regions show increased activation when actions are primarily informative — when the goal is to reduce uncertainty rather than to achieve reward. The "curiosity signal" is neurally distinct from the "reward signal."

Trade-off monitoring: The ACC monitors the trade-off between exploiting known rewards (pragmatic) and exploring uncertain options (epistemic). The exploration-exploitation dilemma is resolved by EFE — policies that simultaneously achieve goals and reduce uncertainty are preferred.

3. Dopamine and Policy Precision

Dopamine plays a critical role in Active Inference through its effect on **policy precision** — the confidence with which the agent selects among policies:

Tonic dopamine: Sets the baseline precision of policy selection. Higher tonic dopamine → more precise (confident) policy selection → more decisive behavior. Lower tonic dopamine → flatter policy distribution → more random behavior (as in Parkinson's disease).

Phasic dopamine: Signals reward prediction errors — the difference between expected and received reward. These RPEs update value estimates and refine future policy evaluations. This is consistent with Schultz's classical work on dopaminergic signaling.

The gamma parameter: In Active Inference, the precision of policy selection is controlled by the parameter γ (gamma), which maps onto dopaminergic function. High γ = more exploitation (confident selection of the best policy). Low γ = more exploration (wider distribution over policies).

Clinical implications: Schizophrenia (elevated dopamine) → overconfident policy selection, jumping to conclusions, aberrant salience. Parkinson's (depleted dopamine) → indecisive, difficulty initiating action, apathy.

4. Working Memory as Active Inference

Working memory — holding information "in mind" — is modeled as **active maintenance of beliefs** about hidden states that are currently unobserved:

Persistent neural activity: Neurons in dlPFC maintain sustained firing during delay periods — this corresponds to maintaining a posterior distribution over hidden states when sensory evidence is absent.

Theta-gamma coupling: Working memory content is encoded in gamma oscillations phase-locked to frontal theta. Each item in working memory occupies a separate gamma burst within the theta cycle. This implements the maintenance of multiple beliefs simultaneously.

Capacity limits: Working memory capacity (~4 items) corresponds to the number of gamma bursts that can fit within one theta cycle — a biophysical constraint on how many posterior distributions can be simultaneously maintained.

5. Neural Basis of Temporal Depth

Planning requires evaluating policies over multiple future time steps. The brain implements this through:

Hippocampal replay: During rest and decision points, the hippocampus replays past trajectories at accelerated speed (sharp wave-ripples). This implements mental simulation — evaluating candidate policies by "running them forward" in experience.

Prospective coding: Hippocampal place cells represent not only current location but upcoming locations — "looking ahead" along planned routes. This neural phenomenon directly implements the T-step policy evaluation in Active Inference.

Prefrontal-hippocampal coupling: During planning, prefrontal theta synchronizes with hippocampal theta, linking abstract goals with concrete spatial trajectories.

Summary

The prefrontal cortex implements hierarchical policy evaluation, with dlPFC maintaining candidate policies, vmPFC computing pragmatic value, ACC monitoring decision difficulty, and frontopolar cortex handling counterfactual reasoning. EFE is neurally decomposed into reward (ventral circuits) and information-seeking (dorsal circuits). Dopamine controls policy precision (decision confidence). Working memory maintains beliefs as sustained neural activity, and hippocampal replay supports temporal planning.

Further Reading

- Friston, K. J. et al. (2017). Active inference, curiosity and insight. *Neural Computation*, 29(10), 2633-2683.
- Schwartenbeck, P. et al. (2015). Optimal inference with suboptimal models. *PNAS*, 112(33), 10368-10373.
- Parr, T. & Friston, K. (2017). Uncertainty, epistemics and active inference. *Journal of the Royal Society Interface*, 14(136), 20170376.